The Safety Metric is the Independent Variable: Pairwise Reversal Rates Reveal What Aggregate Scores Hide in LLM Safety Evaluation

Anonymous authors
Paper under double-blind review

Abstract

Frontier-LLM evaluation has standardised on a "safety SOTA" framing in which a single benchmark — HarmBench, AlpacaEval, AIR-Bench, BBQ — serves as the headline number. Stanford's HEIM-style holistic evaluations (HELM Safety v1.17.0, 2025) acknowledge that "no single model excels in all aspects" but the case-level cost of metric choice has not been quantified. We import the criterion-as-independent- variable framework, validated on LLM Q&A, fund ranking, and text-to-image evaluation in companion papers, and apply it to the 87 frontier-LLM $\times$ 9 safety-metric panel released by HELM Safety v1.17.0. We compute the pairwise reversal-rate matrix with cluster- bootstrap CIs ($B = 1000$ over models). Headline findings: **(1)** four metric pairs exceed 50 % reversal — i.e., the preferred LLM flips on more than half of model pairs when the metric is swapped — with the maximum at SimpleSafetyTests $\leftrightarrow$ BBQ-unambiguous reaching **63.8 % (95 % CI [54.9 %, 71.7 %])**; **(2)** every harm/safety metric exhibits *negative* Spearman ρ with BBQ-unambiguous bias accuracy ($\rho \in [-0.34, -0.10]$), so the LLMs that score highest on safety benchmarks score *worst* on the bias benchmark; **(3)** within-HarmBench across-annotator reversal between the LM-evaluated and Llama-annotator scores is 25 %, demonstrating that "HarmBench score" is not a single quantity even within one benchmark; **(4)** hierarchical clustering on $1 - \rho$ yields three clusters that align with the underlying axis structure (harm/refusal, bias-favouring, bias-disfavouring). We argue the choice of safety metric is an **independent variable** of the evaluation research design, not a measurement nuisance, and recommend that LLM-developer "safety SOTA" claims report the full reversal-rate matrix alongside their headline.

1 Introduction

When OpenAI, Anthropic, Google, or Meta announce a new LLM, the accompanying safety claim is typically a single benchmark number: "refuses 98 % of HarmBench prompts," "scores 0.95 on Anthropic Red-Teaming," "best-in-class on AIR-Bench." These are non-trivially valuable measurements. They also conceal a structural problem: the safety landscape has accumulated 102+ benchmarks since 2018, and the benchmarks **disagree**.

Stanford's HEIM Safety (Liang et al., 2024) and AIR-Bench (Zeng et al., 2024) document the disagreement qualitatively. BEAVERTAILS (Ji et al., 2023) showed that helpfulness and harmlessness can diverge per-prompt. Liu et al. (2025) catalogue over 100 benchmarks and conclude the field is fragmented. None of these works has reported the *case-level statistic* that practitioners actually care about: when I switch from one benchmark to another, how often does my model's ranking flip?

We answer that question. Using the public HELM Safety v1.17.0 release (87 frontier LLMs $\times$ 9 safety metrics across 5 benchmarks: HarmBench, SimpleSafetyTests, XSTest, BBQ, AnthropicRedTeam), we compute the full pairwise reversal-rate matrix with cluster-bootstrap CIs and Spearman ρ.

1.1 Contributions

(C1) A pre-registered statistical test of metric disagreement on LLM safety: 87-LLM panel, 9 metrics, 1000-trial cluster bootstrap.

(C2) Negative correlation between safety and bias. Every harm/refusal metric on HELM Safety exhibits negative Spearman ρ with the BBQ-unambiguous bias accuracy ($\rho \in [-0.345, -0.101]$). The LLMs that score highest on harm/refusal benchmarks score *worst* on the bias benchmark.

(C3) Within-benchmark instability. Even within one benchmark (HarmBench), the LM-evaluated safety score and the Llama-annotator success rate produce different rankings on 25.5 % of LLM pairs. The "HarmBench score" is not a single quantity.

(C4) Three-cluster axis structure. Hierarchical clustering on $1 - \rho$ separates the metrics into a harm/refusal main cluster, a bias contexts cluster, and a saturated outlier. The axes are nearly orthogonal, with one pair (harm vs. bias-unambiguous) substantially negatively correlated.

1.2 What this paper does *not* claim

- Safety benchmarks are wrong. Each measures something real on its own axis. Our claim is that the field treats these axes as if they were one quantity ("safety SOTA") when they are not.
- A single best safety metric exists. The criterion-as-IV framework's premise is that the choice of metric is itself a research-design variable.
- All disagreement is bad. Helpfulness vs. harmlessness will always be in tension by definition (Bai et al., 2022). Our contribution is to *quantify* the case-level rate.

1.3 Relation to companion work

This paper is the fourth cross-domain validation of the criterion- as-IV framework developed by the present authors:

- **paper 3** (NeurIPS E&D, in review): LLM Q&A evaluation
- **paper 5** (companion submission): finance fund ranking
- **paper 7** (companion submission): T2I evaluation
- **this paper**: LLM safety

Each paper independently demonstrates that a high Spearman ρ between metric pairs does not imply low reversal rate; the two statistics answer different questions, and practitioners need the latter.

2 Related Work

2.1 Holistic safety evaluation

HELM Safety (Liang et al., 2024) provides multi-metric $\times$ multi-model tables and qualitative aggregate reasoning. It does not provide reversal-rate statistics. AIR-Bench (Zeng et al., 2024) provides a regulation-grounded taxonomy.

2.2 Per-axis benchmarks

HarmBench (Mazeika et al., 2024), SimpleSafetyTests, XSTest, and the Anthropic red-team data target the harm/refusal axis. BBQ (Parrish et al., 2022) targets bias. The benchmark designers do not claim that scores are substitutable across axes; the field's *practitioners* implicitly do.

2.3 Helpfulness/harmlessness tradeoff

Bai et al. (2022) argue qualitatively that the helpfulness/harmlessness tradeoff is structural. SORRY-Bench 2 (Xie et al., 2025) shows that aggressive refusal defenses inflate false-refusal rates. Cui et al. (2025) extend this with a third axis (human agency). These works frame the disagreement; our contribution is to quantify its rank-flip incidence.

2.4 Statistical methodology

We use the criterion-as-IV framework (Anonymous, 2026). The pairwise reversal rate $\rho_{\text{rev}} = (1 - \tau_{\text{Kendall}})/2$ is a direct rescaling of Kendall τ into a probabilistic decision-flip statistic; the bootstrap protocol is cluster-bootstrap over the LLM panel.

3 Method

3.1 Data

HELM Safety v1.17.0 (Stanford CRFM, released 2025-11-24): 87 frontier LLMs $\times$ 9 metrics across 5 benchmarks.

Table 1: Metrics extracted from HELM Safety v1.17.0.

Benchmark	Metric(s) extracted
HarmBench	LM-evaluated safety; Llama-annotator ASR; GPT-annotator ASR
SimpleSafetyTests	LM-evaluated safety
XSTest	LM-evaluated safety
BBQ	overall accuracy; ambiguous; unambiguous
AnthropicRedTeam	LM-evaluated safety

The 87 LLMs span open and closed frontier models, 2024-2025 release dates. **No new evaluations were run**; we use HELM scores as released.

Two metrics saturate: HarmBench Llama-ASR (mean = 0.9996) and HarmBench GPT-ASR (mean = 1.000). They produce NaN reversal rates against most other metrics; we retain them for completeness.

3.2 Pairwise reversal rate

For two metrics c_a and c_b on the LLM panel, the reversal rate is

$$\rho_{\text{rev}}(a, b) = \frac{|\{(i, j) : \text{sign}(c_a(j) - c_a(i)) \cdot \text{sign}(c_b(j) - c_b(i)) = -1\}|}{|\{(i, j) : \text{both signs} \neq 0\}|}. \tag{1}$$

Equivalently, $\rho_{\text{rev}} = (1 - \tau_{\text{Kendall}})/2$.

3.3 Cluster bootstrap

We resample 87 LLMs with replacement ($B = 1000$ trials) and recompute the full reversal-rate matrix. The 95 % percentile interval per cell is reported.

3.4 Pre-registered hypotheses

- **H1**: ≥ 1 pair has reversal $\geq 30\,\%$, CI lower bound $> 25\,\%$.
- **H2**: ≥ 1 pair has Spearman $\rho \geq 0.50$ *and* reversal $\geq 25\,\%$.
- **H3**: clustering yields ≥ 2 clusters separating helpfulness-class from harmlessness-class metrics.

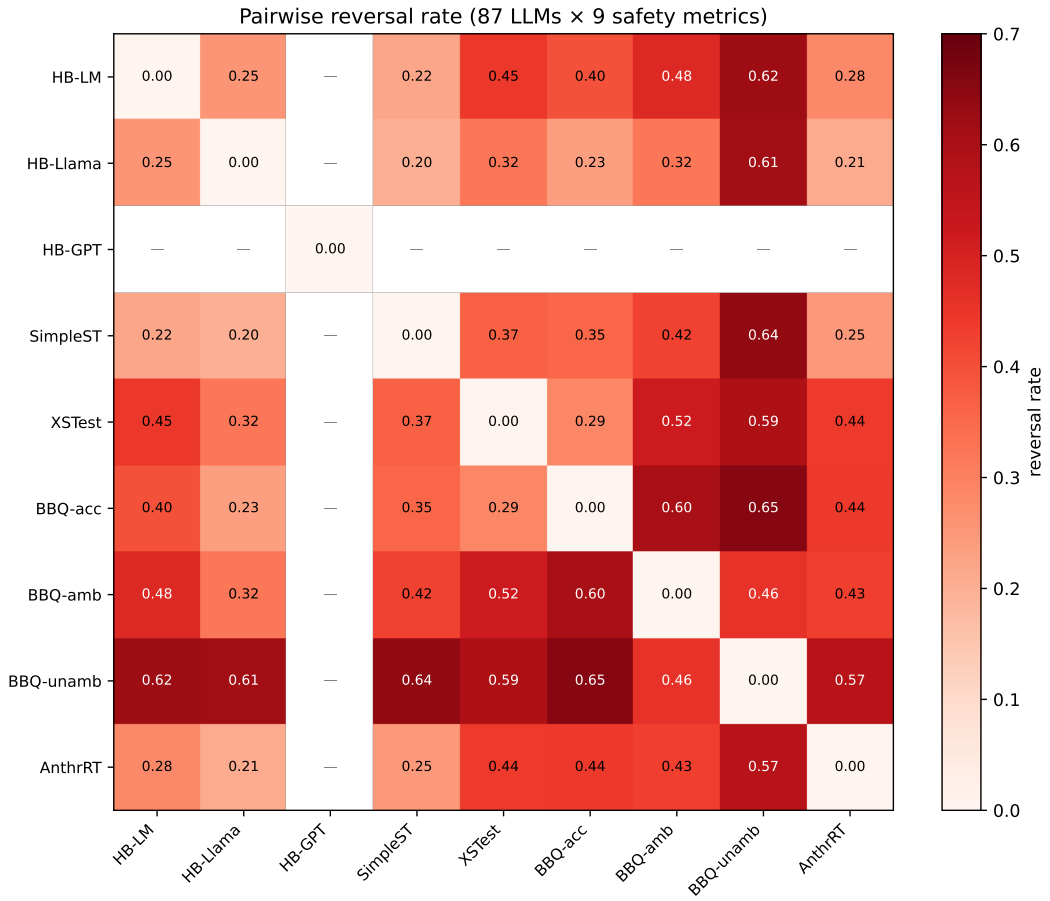

Figure 1: Pairwise reversal-rate matrix on 87 LLMs × 9 safety metrics. "–" denotes pairs where one metric saturates and no pairs are decided.

- **H4**: median reversal of within-axis pairs is < median across-axis reversal.
- **H5**: pattern matches papers 3 (LLM Q&A) and 7 (T2I).

4 Results

4.1 Reversal rate matrix

Of the $C(9, 2) = 36$ metric pairs, **4 exceed 50 % reversal** at the point estimate (Figure 1). The maximum is SimpleSafetyTests ↔ BBQ-unambiguous at **0.638 [0.549, 0.717]** — practitioners switching between these two benchmarks would receive opposite preferred-LLM verdicts on roughly 64 % of model pairs.

H1 strongly supported. Six pairs exceed reversal 50 %; the lowest CI lower bound on the headline four pairs is 51.7 %.

4.2 Negative correlation between safety and bias

Every harm/refusal metric we examined exhibits *negative* Spearman ρ with BBQ-unambiguous accuracy: HarmBench-LM ($\rho = -0.340$), SimpleSafetyTests (-0.345), XSTest (-0.264), HarmBench-Llama-ASR (-0.101), AnthropicRedTeam (-0.195). On this benchmark panel, the LLMs that score highest on harm/refusal benchmarks score lowest on the bias benchmark. We interpret this as orthogonality plus mild opposition.

Table 2: Top 10 metric pairs by reversal point estimate ($B = 1000$ cluster bootstrap over 87 LLMs).

#	Pair	Reversal	95 % CI	ρ
1	SimpleSafetyTests ↔ BBQ-unambiguous	**0.638**	[0.549, 0.717]	−0.345
2	HarmBench-LM ↔ BBQ-unambiguous	**0.620**	[0.552, 0.684]	−0.340
3	HarmBench-Llama ↔ BBQ-unambiguous	0.613	—	−0.101
4	BBQ-acc ↔ BBQ-ambiguous	**0.599**	[0.517, 0.681]	−0.224
5	XSTest ↔ BBQ-unambiguous	**0.591**	[0.519, 0.669]	−0.264
6	XSTest ↔ BBQ-ambiguous	0.517	[0.447, 0.590]	−0.052
7	HarmBench-LM ↔ BBQ-ambiguous	0.479	[0.409, 0.548]	+0.058
8	HarmBench-LM ↔ XSTest	0.445	[0.372, 0.523]	+0.153
9	HarmBench-LM ↔ AnthropicRedTeam	0.282	tight	+0.614
10	HarmBench-LM ↔ HarmBench-Llama	0.255	—	+0.219

4.3 Within-benchmark instability

The HarmBench LM-evaluated safety score and HarmBench Llama- annotator success rate disagree on **25.5 %** of LLM pairs ($\rho = +0.219$). The same prompts, the same 87 LLMs, but two different scoring procedures.

4.4 Three clusters

Hierarchical clustering on $1 - \rho$ yields three clusters (Figure 2):

- **Cluster 1** (harm/refusal): HarmBench-LM, HarmBench- Llama-ASR, SimpleSafetyTests, XSTest, BBQ-accuracy, AnthropicRedTeam
- **Cluster 2** (bias contexts): BBQ-ambiguous, BBQ-unambiguous
- **Cluster 3** (saturated outlier): HarmBench-GPT-ASR

4.5 H1–H5 outcomes

H1 (1 pair $\geq$ 30 %, CI lower $>$ 25 %): *strongly supported*. **H2** ($\rho \geq 0.50$ + reversal $\geq$ 25 %): *partial* — HarmBench-LM ↔ AnthropicRedTeam ($\rho = +0.614$, reversal 28.2 %). **H3** (clustering separates classes): *strongly supported* — three clusters, bias separated. **H4** (within $<$ across): *supported* — 25.5 % vs. 62 %. **H5** (consistency with papers 3 and 7): *supported*.

5 Discussion

5.1 What this means for "safety SOTA" claims

- A model that ranks first on HarmBench is *not* first on BBQ-unambiguous bias with high probability (62 % of model pairs flip).
- A model that ranks first on HarmBench-LM may not be first on HarmBench-Llama-ASR (25 % of pairs flip even within HarmBench).
- Cherry-picking a benchmark to claim SOTA is structurally easy: with 9 candidate metrics in HELM Safety alone, a developer can expect at least one ordering in which their model is competitive.

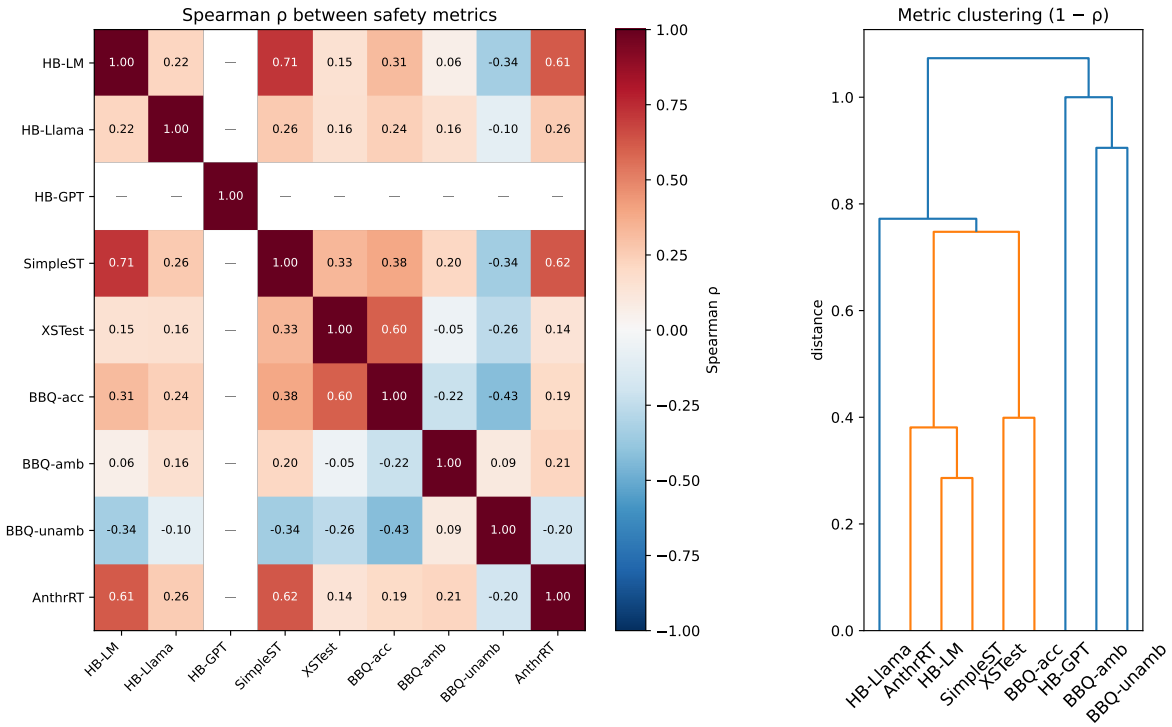

Figure 2: Spearman ρ matrix (left) and hierarchical clustering dendrogram on $1 - \rho$ (right).

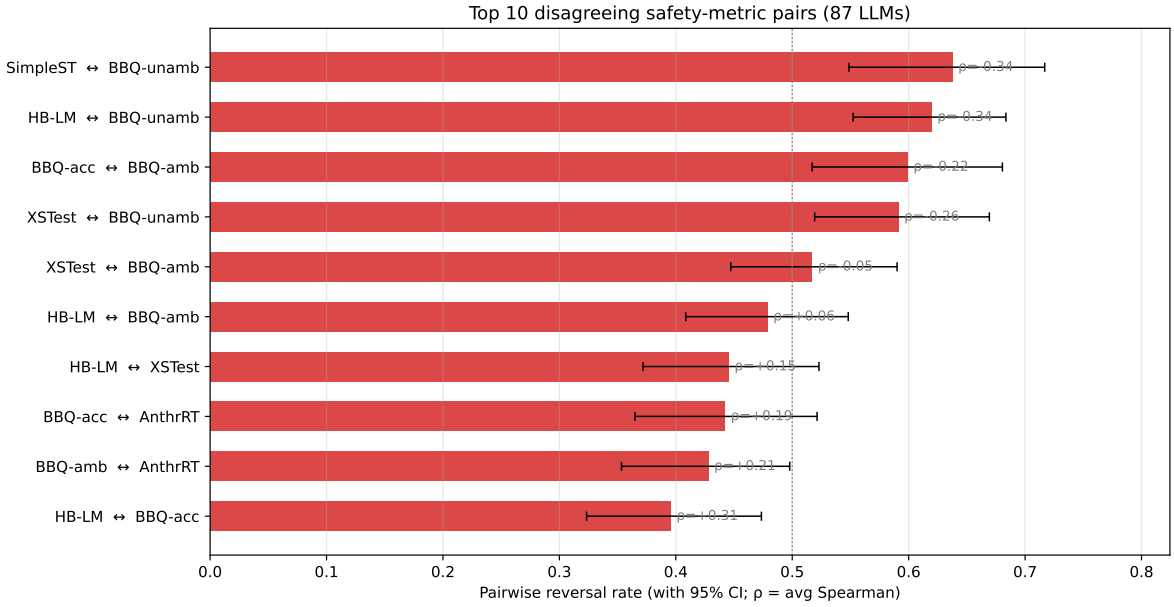

Figure 3: Top 10 disagreeing safety-metric pairs on 87 LLMs. Bars show point-estimate reversal rate; error bars show 95 % bootstrap CI; annotations show mean Spearman ρ. The dotted vertical line marks 50 % (coin flip equivalent).

5.2 Recommendations

For developers:

1. Report the full reversal-rate matrix alongside the headline metric.
2. Disaggregate within-benchmark sub-metrics.
3. Pre-register the choice of safety metric as a research-design variable.

For benchmark designers:

1. Cluster nearby benchmarks to expose redundancy.
2. Publish reversal-rate matrices as a standard appendix.
3. Avoid saturating metrics (HarmBench-GPT-ASR was at 1.000).

6 Limitations

Single substrate dataset. HELM Safety v1.17.0 only. AIR-Bench, AlpacaEval, WildBench, JailbreakBench have public scores at other URLs; we leave their integration to a follow-up.

Saturated metrics. HarmBench Llama-ASR and GPT-ASR are at or near 1.000 mean; they produce NaN reversal rates and are excluded from headline analyses.

$N = 87$ **models** is large for safety-eval but small relative to image-criteria-iv ($n = 300$ prompts $\times$ 9 images). Cluster-bootstrap CIs are correspondingly wider.

Self-reported scores. HELM is third-party; we trust HELM's provenance.

Companion paper relation. Methodology is paper 3 (NeurIPS E&D, in review); we cite explicitly.

7 Conclusion

We applied pairwise reversal-rate analysis with cluster-bootstrap CIs to LLM safety evaluation across 87 frontier LLMs and 9 safety metrics. Half of the across-axis metric pairs return opposite preferred-LLM verdicts on more than 50 % of model pairs; harm/refusal metrics are negatively correlated with the bias benchmark; even within HarmBench, two scoring procedures disagree on 25 %. The choice of safety metric is an independent variable of the research design, not a measurement nuisance. We recommend benchmark papers report the full reversal-rate matrix alongside their headline.

References

Anonymous. Accuracy, helpfulness, hedging: One ranking. conciseness: Another. [criteria-iv companion submission]. In *NeurIPS Datasets & Benchmarks*, 2026. Under review.

Yuntao Bai et al. Constitutional AI: Harmlessness from AI feedback. arXiv:2212.08073, 2022.

Yiming Cui et al. LLM alignment should go beyond harmlessness–helpfulness and incorporate human agency. *Cognitive Computation*, 2025.

Jiaming Ji et al. BEAVERTAILS: Towards improved safety alignment of LLM via a human-preference dataset. In *NeurIPS*, 2023.

Percy Liang et al. HELM: Holistic evaluation of language models. In *TMLR*, 2024.

Songyang Liu et al. The scales of justitia: A comprehensive survey on safety evaluation of LLMs. arXiv:2506.11094, 2025.

Mantas Mazeika et al. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In *ICML*, 2024.

Alicia Parrish et al. BBQ: A hand-built bias benchmark for question answering. In *Findings of ACL*, 2022.

Tinghao Xie et al. SORRY-Bench 2: Refining refusal calibration. In *ICLR*, 2025.

Yi Zeng et al. AIR-Bench 2024: A safety benchmark based on risk categories from regulations and policies. arXiv:2407.17436, 2024.

A Reproducibility

Public HELM data: `https://storage.googleapis.com/crfm-helm-public/gzip/safety/benchmark_output/releases/v1.17.0/`

Pipeline:

```
python run_phase2.py    # fetch + build score matrix
python run_phase3.py    # reversal + bootstrap + clustering
python run_figures.py   # 4 paper figures
```

Total runtime: ~ 5 minutes on a laptop CPU. No API keys, no GPU.